

Multi-Asset Next-Day Return Prediction — Bitcoin, Gold, Silver

Final Year Project · Executive Summary for the Supervisor · Alyan Shahid
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This project is a machine-learning system that forecasts the **next trading day's return** of three assets — Bitcoin (BTC-USD), Gold (GC=F futures) and Silver (SI=F futures) — and presents the forecast, together with its measured reliability, in a web dashboard and a REST API. The forecast target is the next-day log return $y_t = \ln(P_{t+1}/P_t)$; the price shown to the user is recovered as $P_t \cdot \exp(\hat{y})$. Daily data from Yahoo Finance (2018-01-01 to 2026-09-12; Bitcoin 3,147 usable days, Gold/Silver 2,156 exchange days) is enriched with macro series (US Dollar Index, WTI crude, 10-year Treasury yield, S&P 500, VIX) and the crypto Fear & Greed index.

Methodology. A frozen chronological split (train $\leq$ 2025-09-10, validation $\leq$ 2026-02-17, test = 2026-02-18 $\rightarrow$ 2026-09-12) with no shuffling; a scaler fitted on training rows only; 28–29 stationary, backward-looking features; hyper-parameters and the served model chosen by 4-fold expanding-window walk-forward validation inside train+val; and a single evaluation of the untouched test set. Ten model families are compared on identical days: a random-walk baseline, historical mean, ARIMA, Ridge, Random Forest, LightGBM, CatBoost, GRU, LSTM and a stacked ensemble. Twelve specific data-leakage checks are enforced by a 31-case test suite.

Result. The validation-selected models are gradient-boosted trees (CatBoost for Bitcoin, LightGBM for Gold and Silver). On the untouched test set, Bitcoin and Gold remain at the random-walk floor: return-RMSE within ± 0.5 % of the naive forecast (Diebold–Mariano $p = 0.42\text{--}0.74$), directional accuracy 47–51 %. **Silver is the exception:** 62.2 % directional accuracy on 143 unseen days (binomial $p = 0.002$), $R^2 = +0.044$ in return space and a significantly lower squared error than the random walk (DM $p = 0.002$), consistent across every volatility and trend regime of the window — but its walk-forward figure is 52.8 %, so the effect is period-specific and reported as such. Overall the results are consistent with weak-form market efficiency; the application reports the measured error next to every forecast.

Improvement over the initial version. An audit of the original codebase found a price-reconstruction bug that corrupted every dollar metric, 30 % fabricated weekend rows for the metals, hyper-parameters selected on the test set, early stopping on training data, non-stationary price-level features, a served ensemble that predicted a constant, a broken inference path and misleading R^2 values (≈ 0.95 on price levels, which a random walk also achieves). All were fixed. A second, measured improvement phase then added 2.9× more training data (2018 $\rightarrow$), volatility/momentum features, re-tuning, and logged design experiments: on identical unseen days the served models improved by 1.9 % (Gold) and 2.8 % (Silver) in return-RMSE and were unchanged for Bitcoin.

Status. Working Streamlit dashboard (live forecast with uncertainty band, a *Predict-a-Day* demonstration on the unseen test period that reveals the actual price and the error, a full prediction history, validation/test tables, regime analysis and experiment tables) and FastAPI endpoints; reproducible pipeline (make pipeline); documentation for methodology, features, results, limitations, demonstration and viva.

Key facts

Item	Final implementation
Assets / horizon	Bitcoin, Gold (GC=F), Silver (SI=F); next trading day
Target	log return $\ln(P_{t+1} / P_t)$; price = $P_t \cdot \exp(\hat{y})$
Data	Yahoo Finance daily OHLCV 2018-01-01 $\rightarrow$ 2026-09-12 (BTC 3,147 / metals 2,156 usable days) + DXY, WTI, 10-y yield, S&P; 500, VIX, Fear & Greed

Item	Final implementation
Features	29 (BTC) / 28 (Gold) / 29 (Silver) stationary, backward-looking features incl. HAR/EWMA volatility and momentum; raw price levels never used as inputs
Split	train $\leq$ 2025-09-10 · validation $\leq$ 2026-02-17 · unseen test 2026-02-18 $\rightarrow$ 2026-09-12 (207 / 143 / 143 days), chronological, evaluated once
Validation	4-fold expanding-window walk-forward on train+val for tuning and model selection
Models	Naive random walk, historical mean, ARIMA, Ridge, Random Forest, LightGBM, CatBoost, GRU, LSTM, stacked ensemble
Served model	CatBoost for Bitcoin, LightGBM for Gold and Silver (walk-forward validation winners; early-stopped after 10–50 rounds)
Metrics	MAE/RMSE/MAPE (\$), RMSE/MAE/R ² (return), directional accuracy + binomial p, Diebold–Mariano vs random walk, strategy backtest
Stack	Python 3.9, pandas, scikit-learn, LightGBM, CatBoost, TensorFlow/Keras, statsmodels, Streamlit, FastAPI, Docker, pytest (31 cases), GitHub Actions
Demo	Predict-a-Day on the unseen test period (predicted vs actual vs error), prediction history, live forecast with uncertainty band

Headline test-set results

Asset	Model	MAE (\$)	RMSE (\$)	MAPE %	R ² (return)	Dir. acc. % (p)	DM p vs naive
Bitcoin	Naive (random walk)	1,066.42	1,451.74	1.52	-0.001	—	—
Bitcoin	CatBoost (served)	1,071.08	1,453.76	1.53	-0.004	46.9 (0.83)	0.74
Gold	Naive (random walk)	58.26	75.76	1.29	-0.002	—	—
Gold	LightGBM (served)	58.02	75.30	1.29	+0.008	51.0 (0.43)	0.42
Silver	Naive (random walk)	1.79	2.37	2.49	-0.001	—	—
Silver	LightGBM (served)	1.75	2.32	2.42	+0.044	62.2 (0.00)	0.00

Unseen test period 2026-02-18 $\rightarrow$ 2026-09-12. Full 10-model tables: [results/FINAL_RESULTS.md](#) and the main report.

Reading the numbers

R² is reported on returns, where 0 means "as good as predicting the mean". The original project reported R² $\approx$ 0.95 on price levels — a value the random walk also achieves, which is why it was misleading. For Bitcoin and Gold the served models' errors do not differ significantly from the random walk's (DM p = 0.74, 0.42). For Silver they do (DM p = 0.002; 62 % direction, p = 0.002), consistently across regimes — but the same model scored 53 % on walk-forward validation, so this is a period-specific result, stated with that caveat. On identical unseen days the improvement phase lowered return-RMSE by 1.9 % (Gold) and 2.8 % (Silver) and left Bitcoin unchanged; the gain came from 2.9× more data, not from model complexity.

Current status: implementation complete; pipeline reproducible; dashboard (with the unseen-day demonstration) and API verified running for all three assets; documentation, demo guide and viva notes prepared. **Recommended framing:** a leakage-audited forecasting benchmark and decision-support prototype whose predictions can be verified day by day — not a market-beating predictor.